

My name is Zhan Yongting, and I'm currently a third-year undergraduate majoring in Automation at Harbin Institute of Technology, Shenzhen. I come from Chaozhou in Guangdong Province. I'm an outgoing person, and in my free time, I enjoy playing badminton or the piano. My main research interest is embodied intelligence, Reinforcement learning and so on.

My overall GPA is 3.897 out of 4.0, ranking 7th out of 180 students in my major, and I was awarded the National Scholarship.

I come from a happy family. My father works as a firefighter, and my mother is a middle school teacher. I am an only child, and my parents have always been very supportive of both my academic pursuits and my interests.

My main research interest is embodied intelligence, particularly visual navigation. I worked as a co-first author on a project called MeanFlowNav. The motivation was that existing navigation policies based on diffusion models or flow matching suffer from high inference latency. We proposed modeling the waypoint generation process with an average velocity field instead, which lets us compress the multi-step integration into a single forward pass. We validated this on a four-wheel robot platform we built ourselves, across six real-world scenarios, and we're currently preparing the paper for submission.

In this project, I collected and processed data, trained models, and gained an understanding of how a research project is carried out.

Beyond research, I've been active in robotics competitions. Most notably, I won a National Special Prize at the 19th "Challenge Cup" AI competition, where our team built a ROS-based state-machine planner for a humanoid robot performing warehouse tasks — I was responsible for the motion control using model predictive control, and our solution completed the task with a perfect score in 8 out of 10 repeated trials. I've also worked on embedded systems and control, including a STM32-based two-axis gimbal tracking system using YOLO11 for object detection, and an open-source balance-riding-leg robot project.

Overall, my background combines a solid foundation in control theory and embedded systems with hands-on experience in modern learning-based robotics — and I'm hoping to continue this direction in graduate study, working on embodied intelligence and efficient robot learning.

I am a probationary member of the Communist Party of China.

"I am applying for a graduate program in Control Science and Engineering at the Institute of Automation."

"My campus is located in Shenzhen, in the Guangdong-Hong Kong-Macao Greater Bay Area, close to the university towns area with several other universities nearby. The campus itself is relatively new and modern, with clean, spacious layouts and a lot of green space. Being in Shenzhen, the climate is warm and humid most of the year, and the campus has a fairly international, tech-driven atmosphere overall, since it's surrounded by many high-tech companies and research institutions in the city."

"Shenzhen is a major city in southern China, right next to Hong Kong, in the Guangdong-Hong Kong-Macao Greater Bay Area. It's known as one of China's fastest-growing cities — about 40 years ago it was a small fishing village, and now it's a major hub for technology and innovation, home to companies like Huawei, Tencent, and DJI. The city has a young population, a fast pace of life, and a strong tech-oriented culture, which makes it a great environment for someone studying engineering and robotics like me."

"Beijing is the capital of China, located in the north of the country. It has a long history and served as the capital for several dynasties, so it has a rich historical and cultural heritage — like the Forbidden City, the Great Wall, and Tiananmen Square. It also features modern development as a political, cultural, and technological center. It's also home to many top universities and research institutions, including the Chinese Academy of Sciences, which makes it one of the most important hubs for scientific research in China."

"The Institute of Automation is a national research institute under the Chinese Academy of Sciences. It's one of the earliest and most respected institutes in China for automation and AI research. The campus is small and quiet, more like a research campus than a university. It has a strong academic atmosphere. Inside the institute, there are several national key laboratories, and researchers work closely with graduate students on real projects. Overall, I love the environment here."

"I visited Beijing before as a tourist. I went to the Forbidden City and the Great Wall. The Forbidden City was huge and very impressive. I could feel the history in every building. The Great Wall was also amazing, though it was quite steep to climb. It was also very cold that day, but it was still a lot of fun. Both places left a deep impression on me. I really enjoyed the trip."

"My hobbies are playing the piano and badminton. I've been playing the piano since I was young. I usually play to relax after studying. I also enjoy playing badminton. It helps me stay active and relieves stress. Both hobbies help me balance my life outside of research."

"I started learning the piano in elementary school, and now I'm at Level 7. I like Croatian Rhapsody because it feels really cool to play and it helps me relieve stress."

"One of the biggest challenges I faced was during a competition in the summer after my freshman year. We had been preparing for this competition for several months, but just ten days before the selection round, we discovered that one part of our design actually violated the competition rules. So we had to redesign a new solution and debug it within only ten days. It was very stressful, but in the end we managed to finish it in time, and we won the National First Prize."

"My favorite novel is Little Women. It tells the story of four sisters growing up together. I like how each sister has her own personality and dreams. The story shows the importance of family and self-growth. It's warm, but also inspiring. "

"My hometown is Chaozhou. It's a city in Guangdong Province, in southern China. It has a long history and rich culture. It's especially famous for its food, like Chaozhou beef hot pot and flat rice noodle. It's also known for its traditional culture, like Chaoshan opera and Gongfu tea. It's a comfortable city to live in.."

"I think my strongest advantage is that I'm good at getting to the root of a problem. I calm down and debug carefully, trying to figure out what exactly is causing the experiment to underperform."

"I think my strongest advantage is that my background combines a solid foundation in control theory and embedded systems with hands-on experience in modern learning-based robotics, which help me to do better in my future research project."

"I think the most important ability for a graduate student is independent thinking — the ability to identify a real problem, form your own hypothesis, and design experiments to test it, instead of just following instructions. Research often doesn't have a clear answer, so you need to be comfortable with uncertainty and keep trying different approaches when something doesn't work. I also think persistence matters a lot, because most experiments fail before they succeed. From my own experience working on MeanFlowNav, a lot of the progress came from repeatedly debugging and adjusting the model after failed attempts, not from getting things right the first time."

"One of my weaknesses is that I haven't yet completed a full research project entirely on my own. Another weakness is that sometimes I tend to forget what I've learned relatively quickly, especially if I don't use it often."

Personality

I am outgoing and patient, and I love to make more friends in daily life, uh but sometimes I may get anxious over the things that have not happened

What motivates you most

Describe a moment that you work under stress

What have you learned from the xxx

Future plan \what to do in future five years \职业规划

运动控制代码 motion control code

What can you contribute

My failure

" In the National Undergraduate Electronics Design Contest, we had to finish a project within three days. We struggled with the PID tuning, but kept trying. In the end, we only completed a small part and got a low score."

What have you learned

What skill do you want to get what will you do

Yeah Uh, the project will uh publish its code website, so I can uh fetch the code from the website and to um look into how they realize the function, and I can uh know more about the function of the code and so that I can be more familiar with the coding skill

Why do want you join our project

The difficulty you met in the MFN project and tie it briefly to what you learned about yourself/Describe : solve a tough problem by yourself

"One of the biggest challenges was training a model to predict temporal distance."

"Our early results were weak, so we analyzed the data and expanded the training set. After retraining, performance improved. I learned to Analyze problems systematically and stay patient through iteration."

What can you do in the team (in future team work)

Why should we choose you

my background combines a solid foundation in control theory and embedded systems with hands-on experience in modern learning-based robotics,I can contribute both theoretical understanding and practical coding skills.

Long term plan

I wish to sharpen my research skills and learn more theory about embodied intelligence and reinforcement learning, and in the future ,I would like to go to the industry and apply the theorectical knowledge to the real projects.

"I want to strengthen my grasp of embodied AI and reinforcement learning, especially how to design and debug learning pipelines."